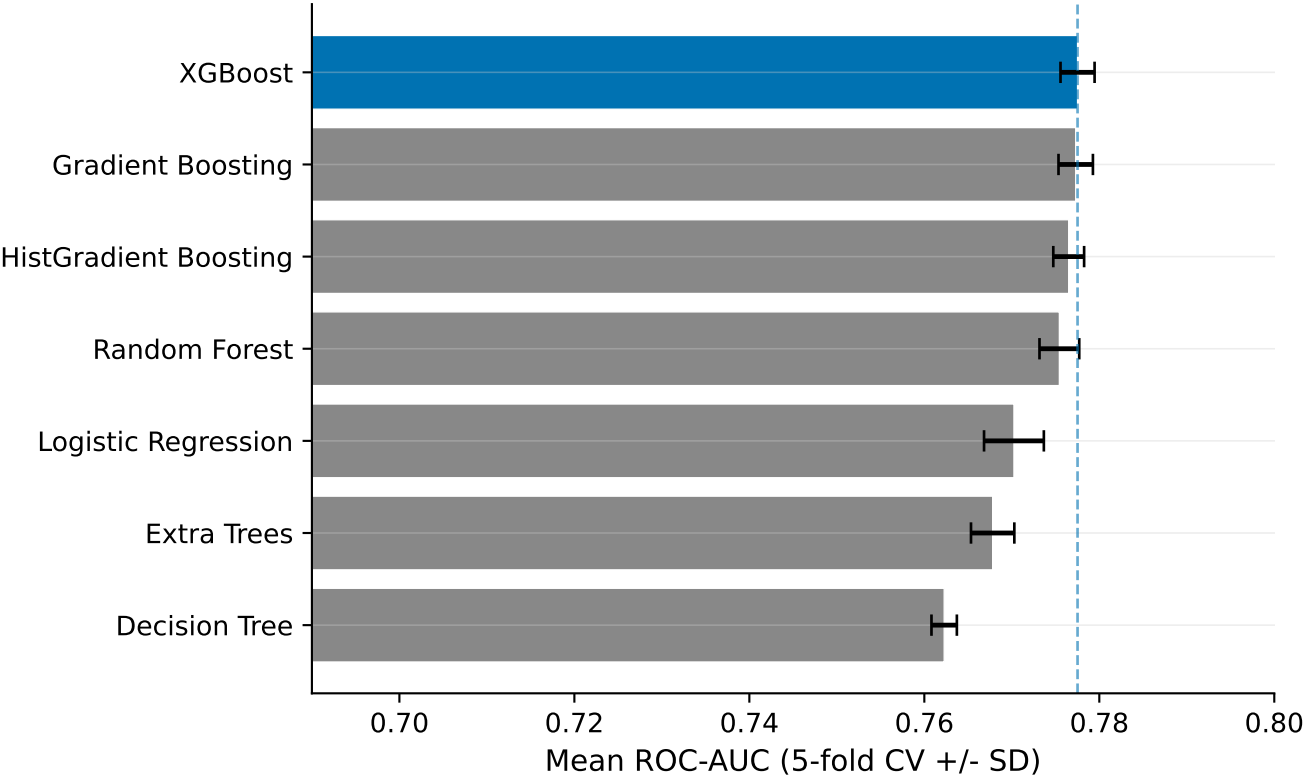
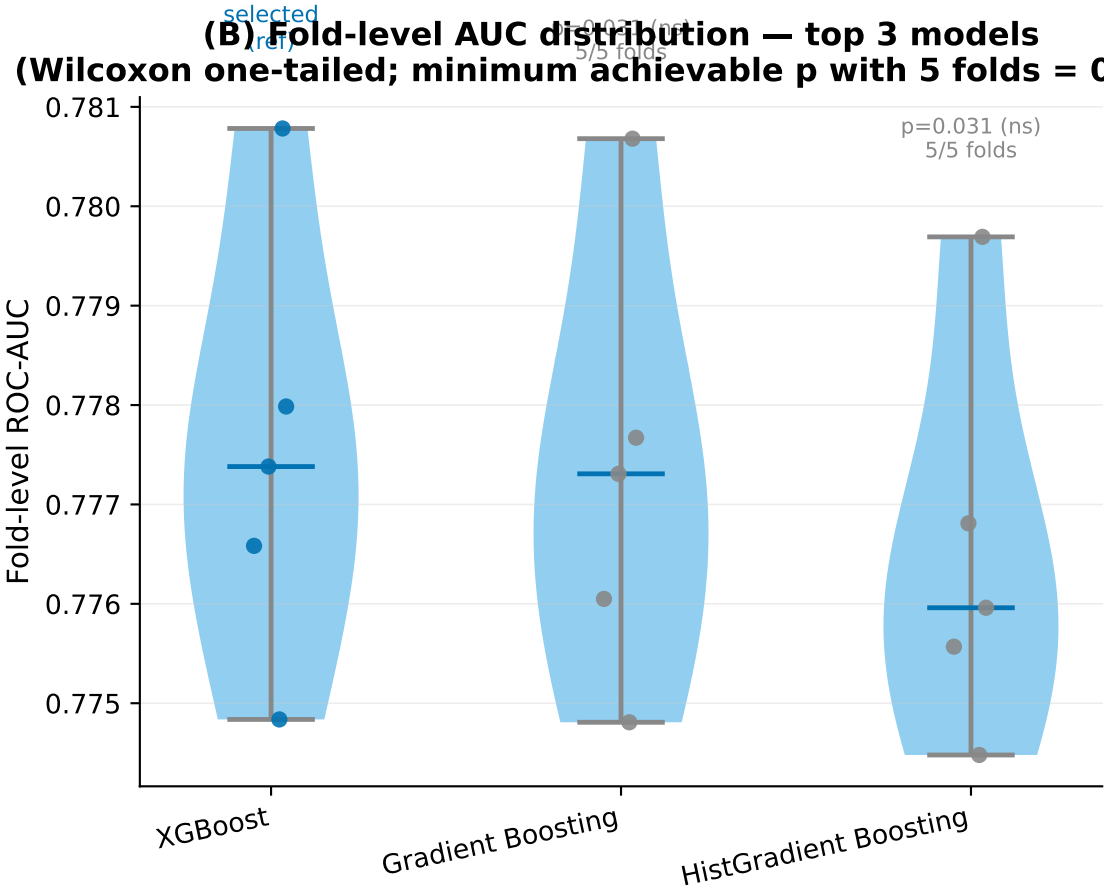


RQ1: Cross-validated model comparison
(XGBoost selected on mean AUC; fold-level differences not statistically significant)

A (A) Model discrimination — mean +/- SD across 5 folds



(B) Fold-level AUC distribution — top 3 models
(Wilcoxon one-tailed; minimum achievable p with 5 folds = 0.0312)



Note (N1): With 5 cross-validation folds the minimum achievable one-tailed Wilcoxon p-value is 0.0312. All comparisons returned p=0.25, confirming no consistent fold-level dominance. XGBoost was selected on highest mean AUC and consistency with the BRFS diabetes screening literature, not statistical superiority.